Lab-6 experiment

on MQTT

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Course: Controllers and Electronics

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Note: In this lab we both team member had performed the experiment of Team1 and Team2 . Additionally, we took the help of Ai tools including ChatGPT and Gemini .

1. Introduction

The objective of this laboratory work was to design, implement, and evaluate a simple Internet of Things (IoT) system using the MQTT (Message Queuing Telemetry Transport) protocol. The system demonstrates real-time communication between a sensor node and an actuator node using lightweight publish–subscribe messaging.

In this experiment, ambient light intensity is measured using a Pmod ALS light sensor connected to a Raspberry Pi . The measured brightness values are published to an MQTT broker. A second Raspberry Pi subscribes to the same topic and controls an LED based on the received data. This setup reflects a typical IoT architecture used in smart environments.

2. System Description

The system has two parts:

2.1Sensor Node (Team 1):

- Measures light using a Pmod ALS light sensor
- Reads the sensor using SPI communication
- Sends brightness values (0–1023) to an MQTT broker every second

2.2 Actuator Node(Team2):

- Receives brightness values using MQTT
- Turns an LED ON or OFF based on the received value
- LED turns ON when brightness is greater than 500
- LED turns OFF when brightness is 500 or less

3.Hardware used:

Following are the hardware which is used by team 1:

- Raspberry Pi Zero 2W
- Pmod ALS light sensor
- Breadboard and jumper wires

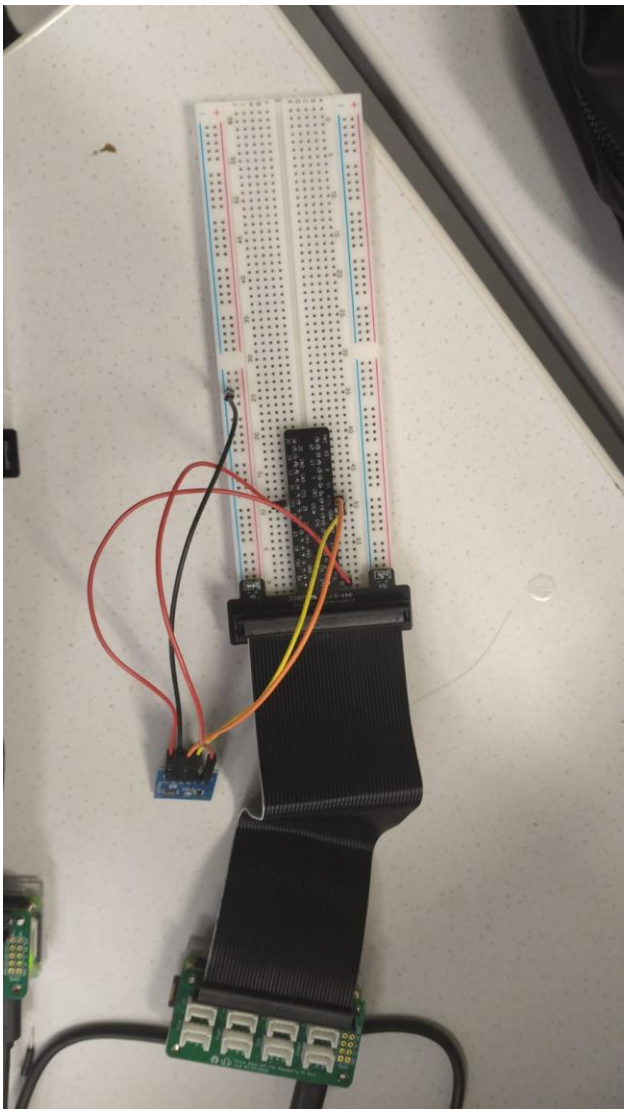

Figure 1. Team 1 hardware setup with Raspberry Pi Zero 2W and Pmod ALS light sensor

Also, the hardware component used by team 2 are present below:

- Raspberry Pi Zero 2W
- LED
- 220 Ω resistor
- Breadboard and jumper wires

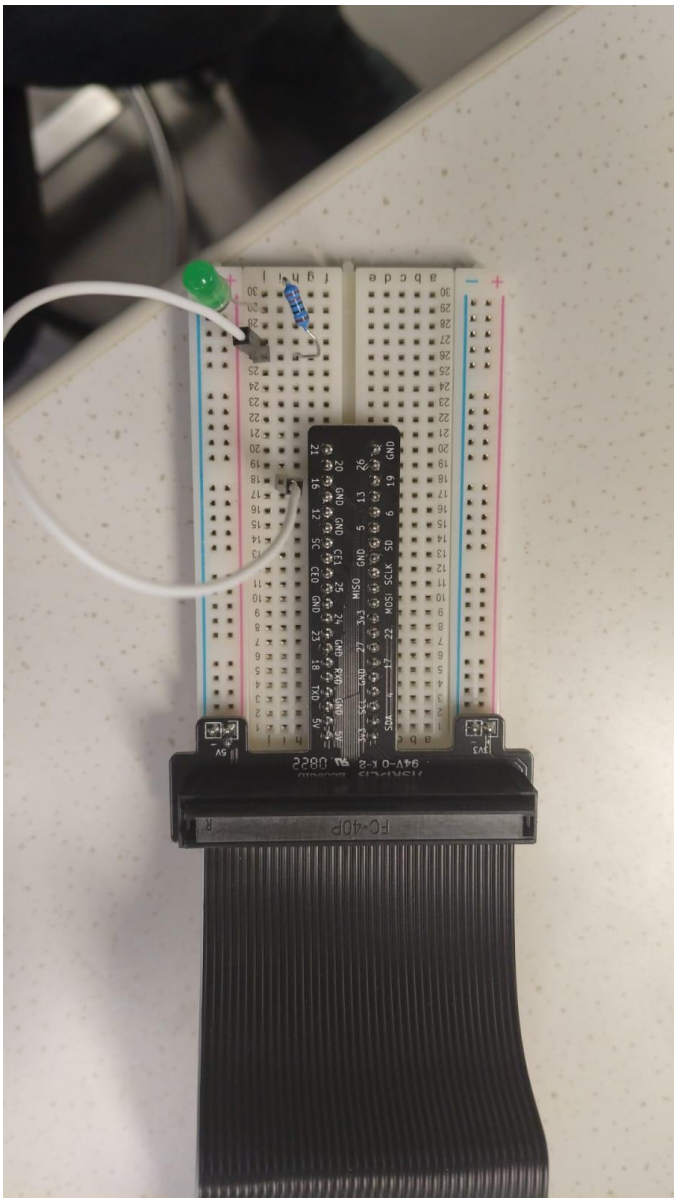

Figure 2. Team 2 hardware setup showing Raspberry Pi Zero 2W controlling an LED

4. Software Overview

Sensor Program (Publisher)

- Reads light values from the sensor
- Sends the values to the MQTT broker
- Publishes data once every second

```
labuser@RiiA103RPiZe29: ~/l
  GNU nano 8.4                                    team_1
import spidev
import time
import paho.mqtt.client as mqtt

BROKER = "10.10.200.15"
TOPIC = "lab6/light"

spi = spidev.SpiDev()
spi.open(0, 0)
spi.max_speed_hz = 500000

client = mqtt.Client()
client.connect(BROKER)

def read_als():
    resp = spi.xfer2([0x00, 0x00])
    value = ((resp[0] & 0x03) << 8) | resp[1]
    return value

while True:
    brightness = read_als()
    print("Brightness:", brightness)
    client.publish(TOPIC, brightness)
    time.sleep(1)

                                  [ Read 24 lines ]
^G Help      ^O Write Out  ^F Where Is   ^K Cut       ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File  ^\ Replace    ^U Paste     ^J Justify    ^/ Go To Line M-E Redo
```

Figure 3. Team 1 Python program for reading light sensor data and publishing it to the MQTT broker

LED Program (Subscriber)

- Receives brightness values from MQTT

- Compares the value to a threshold
- Controls the LED based on the value

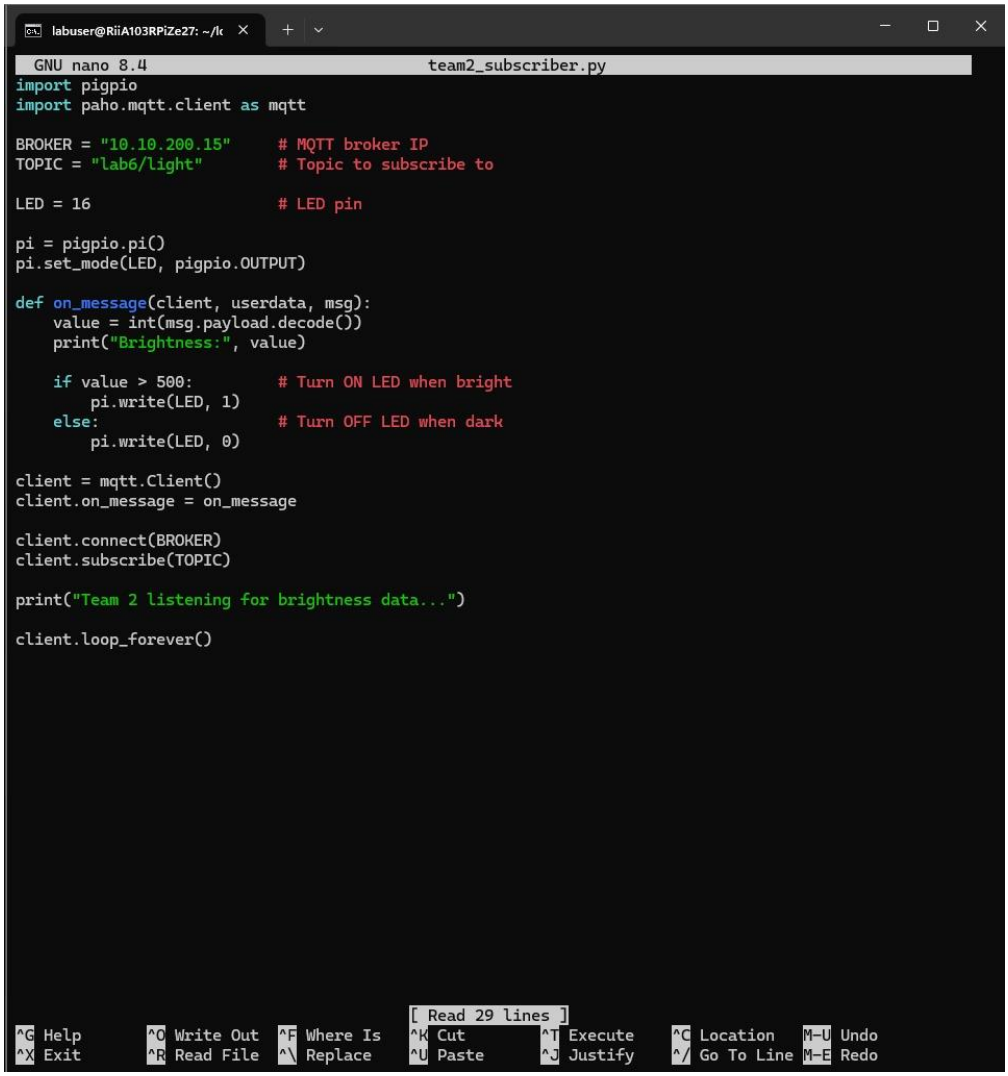

Figure 4. Team 2 Python program subscribing to MQTT data and controlling the LED

5. Results

The MQTT communication worked correctly. The second Raspberry Pi received the same brightness values that were sent by the first Raspberry Pi.

- Low light → LED OFF
- High light → LED ON

When a phone flashlight was pointed at the sensor, the LED turned ON. When the sensor was covered, the LED turned OFF. This shows that the system reacted correctly and in real time.

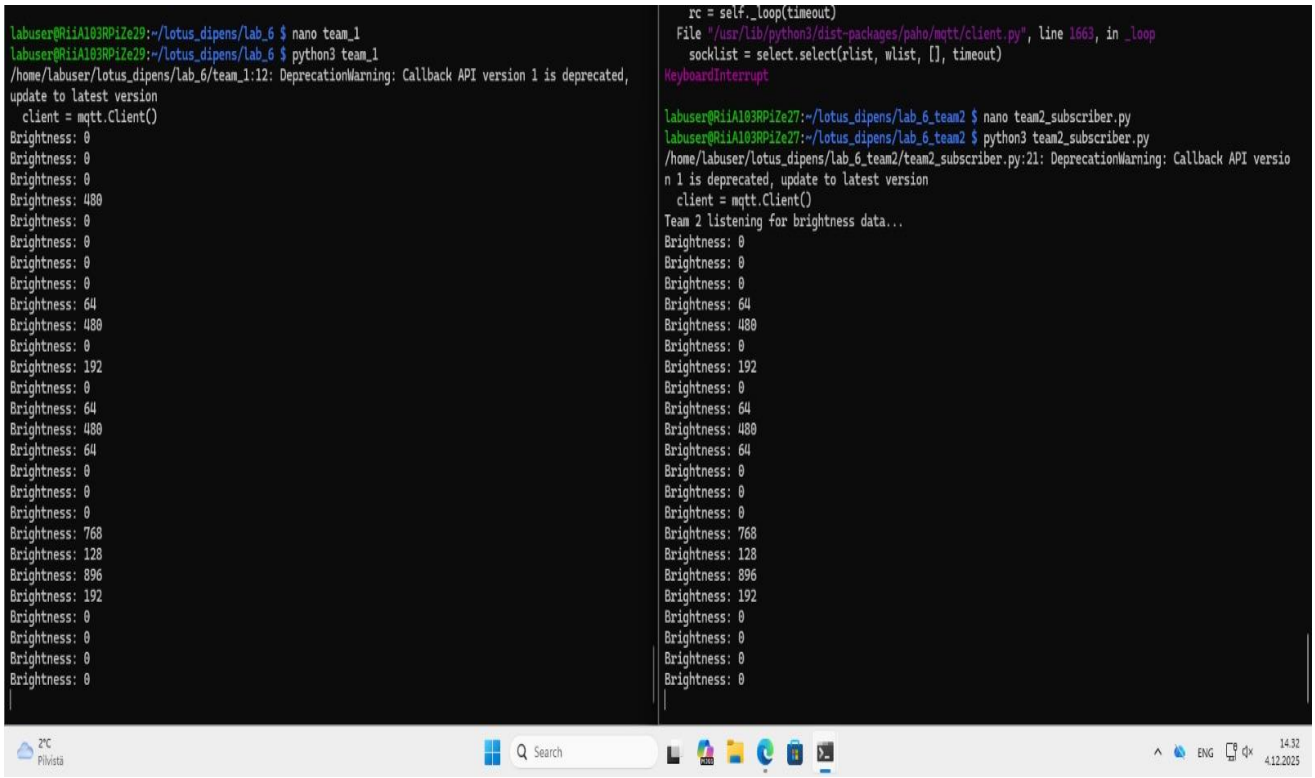

Figure 5. Team 1 and Team 2 programs running simultaneously and exchanging data via MQTT

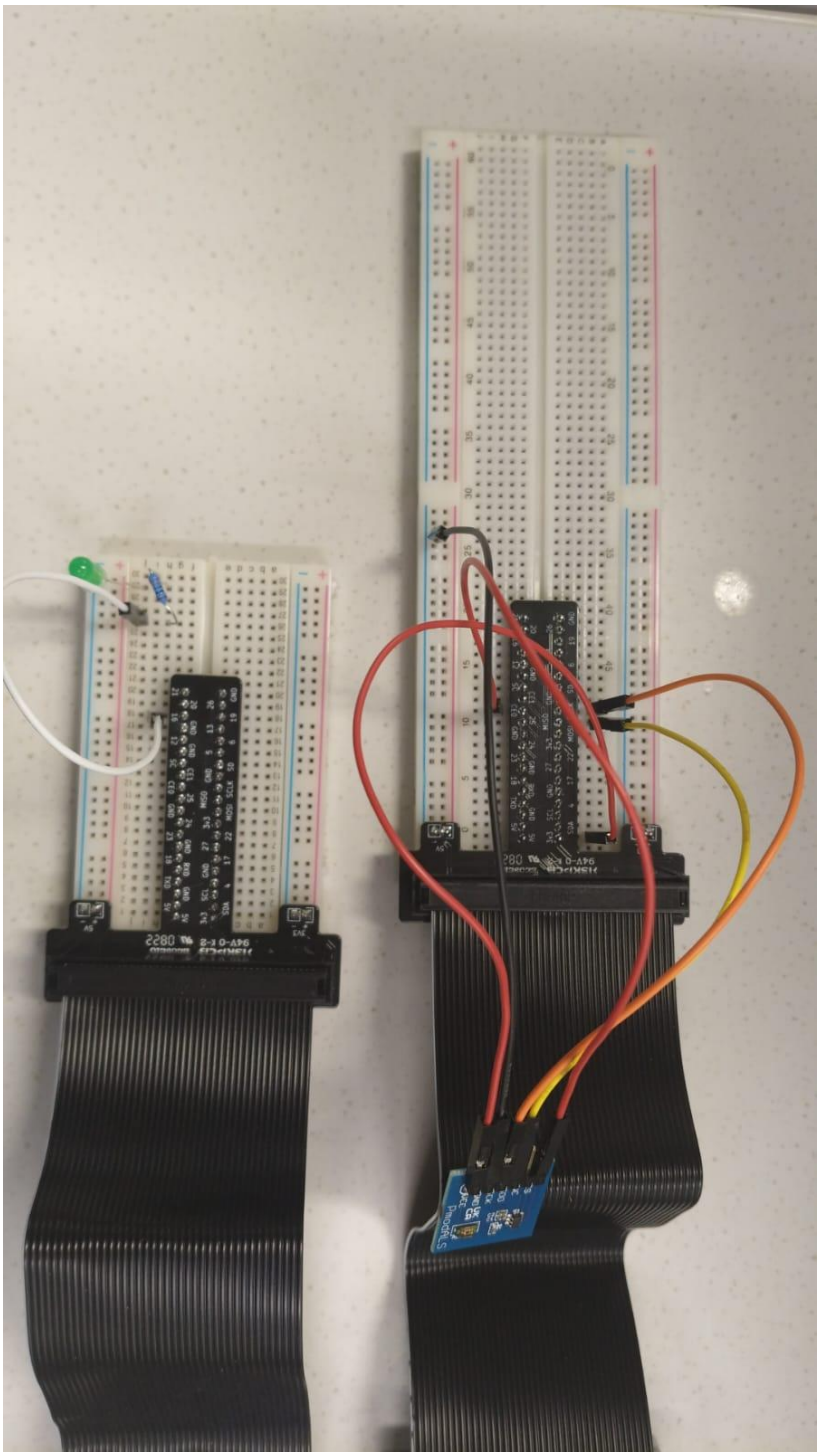

Figure 6. Final system setup showing Team 1 light sensor node and Team 2 LED actuator nod

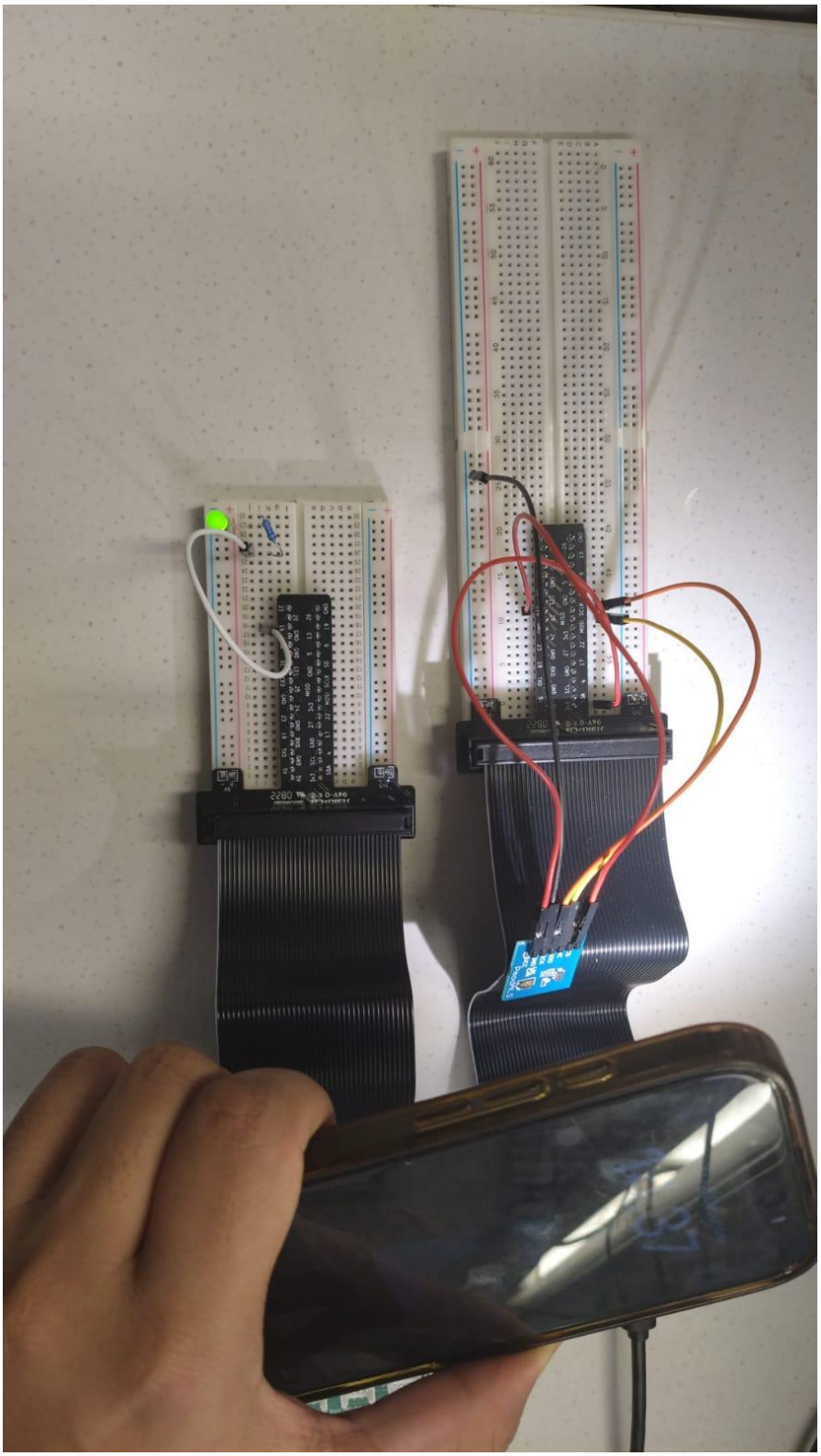

Figure 7: LED activation in response to increased brightness detected by the light sensor

6. Conclusion

In this lab, we successfully built a simple IoT system using MQTT. One Raspberry Pi measured light and sent the data, while another Raspberry Pi received the data and controlled an LED.

As shown in Figure 7, increasing the light intensity using a flashlight caused the LED to turn ON immediately.

The system worked reliably and showed how MQTT can be used to connect sensors and actuators in IoT applications.